INCH-POUND

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MILITARY SPECIFICATION

GRINDING MACHINES, CYLINDRICAL, EXTERNAL, PLAIN, POWER

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers electric motor driven, floor mounted, horizontal spindle, power operated, external, cylindrical, plain grinding machines with equipment and accessories specified herein.

1.2 Classification. The grinding machines shall be of the following styles and the sizes listed in table I. The styles and sizes to be furnished shall be as specified (see 3.5 and 6.2).

Style A - Straight wheelslide

Style B - Angular wheelslide

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Industrial Plant Equipment Center, ATTN: DIPEC-SSM, 2163 Airways Boulevard, Memphis, Tennessee 38114-5051, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-M-18058 - Machinery, Metal and Woodworking, Packaging of.

STANDARDS

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Services.

MILITARY

MIL-STD-461 - Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interferences.

MIL-STD-462 - Electromagnetic Interference Characteristics, Measurement of.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of This document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

CODE OF FEDERAL REGULATIONS (CFR)

U. S. DEPARTMENT OF LABOR, OCCUPATIONAL SAFETY AND HEALTH ADMINISTRATION (OSHA)

29 CFR 1910 - Occupational Safety and Health Standards.

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(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402-0001.)

(Copies of specifications, standards, handbooks, drawings, publications, and other government documents required by the contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN GEAR MANUFACTURERS' ASSOCIATION (AGMA)

AGMA 390.03a - Gear Classification, Materials and Measuring Methods for Unassembled Gears.

AGMA 2000-A88 - Gear Classification and Inspection Handbook.

AGMA 2001-B88 - Fundamental Rating Factors and Calculation Methods for Involute Spur and Helical Gear teeth.

(Application for copies should be addressed to the American Gear Manufacturers' Association, Standards Department, 1500 King Street, Suite 201, Alexandria, VA 22314-2717.)

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI B5.1 - T-Slots, Their Bolts, Nuts and Tongues. (DOD adopted)

ANSI B5.9 - Spindle Noses for Toolroom Lathes, Engine Lathes, Turret Lathes, and Automatic Lathes. (DOD adopted)

ANSI B5.1O - Machine Tapers. (DOD adopted)

ANSI B1l.9 - Safety Requirements for the Construction, Care, and Use of Grinding Machines. (DOD adopted)

ANSI/ISO 54 - Cylindrical Gears For General Engineering and for Heavy Engineering, Modules and Diametral Pitches of.

(Application for copies should be addressed to the American National Standards Institute, ATTN: Sales Dept., 1430 Broadway, New York, NY 10018-3363.)

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JOINT INDUSTRY COUNCIL (JIC)

JIC H-1 Hydraulic Standards for Industrial Equipment and General Purpose Machine Tools.

(Application for copies should be addressed to the Joint Industrial Council. c/o The National Machine Tool Builders' Association, 7901 Westpark Drive, McLean, VA 22101-4269.)

NATIONAL ELECTRICAL MANUFACTURERS' ASSOCIATION (NEMA)

NEMA MG 1 - Motors and Generators. (DOD adopted)

(Application for copies should be addressed to the National Electrical Manufacturers' Association, 2101 L Street, N.W., Washington, DC 20037-1580.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 79 - Electrical Standard for Industrial Machinery.

(Application for copies should be addressed to the National Fire Protection Association, Batterymarch Park, Quincy, MA 02269-9101.)

(Non-Government standards and other publications are normally available from the organizations which prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of Precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), one complete grinding machine shall be subjected to first article inspection (see 6.3), in accordance with 4.4.

3.2 Design. The grinding machine shall be new (not a prototype) and one of the manufacturer's current production models capable of operations in accordance with the requirements herein. The machine shall include all components, parts, and features necessary to meet the performance requirements specified herein. All parts subject to wear, breakage, or distortion shall be accessible for adjustment, replacement, or repair. Style A and style B machines shall have the same operational and performance capabilities, the major design difference being the configuration of the wheelhead slide guideways.

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3.2.1 Measuring and indicating device calibrations. Measuring and indicating devices shall be graduated in inch-pound (see 6.4.1), metric (see 6,4.2), or dual (inch-pound and metric) units as specified by the procuring activity see 6.2). Regardless of the measurement system used, all measuring and indicating devices on the machine shall be graduated in the same system.

3.2.1.1 Dials. All dials used to indicate machine axes position or tool movement shall be graduated in increments not larger than 0.0001 inch. Dial and handwheel circumferences shall be permanently and legibly engraved or etched on a nonglare background with graduations that can be read from the operator's normal position. All feed dials shall have independent zero adjustments and shall be calibrated in such a manner that the last dial graduation progresses into and is continuous with the first dial graduation as the dial is rotated through the zero position.

3.2.1.2 Digital readout system. When specified (see 6.2), a digital readout system shall be installed on the x, y, and z axes of travel. The system shall consist of a position feedback device, an electronic display for each axis, and all related components, mounting hardware, and electrical equipment necessary to ensure complete operation of the system. The system shall provide a readily visible digital display over the full working range of each axis on which installed. The system resolution shall be 0.0005 inch. The system accuracy shall be equal to the system resolution, verified upon installation and calibration of each system on the machine. The repeatability shall be 0.0002 inch, measured over the full travel on each axis. Special requirements for the system components shall be as specified (see 6.2). The display shall indicate axis location in both the inch-pound and metric units. An input source fluctuation of ± 10 percent shall not adversely affect the system operation.

3.2.1.2.1 Position feedback device. A position feedback device including all related components, hardware, and electrical equipment shall be provided for each axis having a readout system. The feedback device shall be of the independent type with a resolution equal to the resolution of the digital readout system. Each feedback device shall be installed with a cover or guard to prevent physical damage to any portion of the feedback device. The connecting cable from the position feedback device shall be electronically shielded to protect against electronic interference and shall be armored to protect against contamination and damage. The device shall provide axis travel feedback over the entire range of axis travel for each specified axis. Installation of the position feedback-device shall not limit machine travel on any axis nor eliminate manual machine stops. Provisions shall be made to prevent damage to the feedback device in the event of overtravel. The feedback unit and related components shall be protected from contamination. Position feedback devices shall be easily and quickly interchangeable between like machines procured under the same contract.

3.2.1 .2.2 Electronic displaY console. The electronic display console for the digital readout shall contain a universal digital display with individual illuminated digits of not less than 0.43 inch in height. Digits shall be high efficiency displays for high light ambient conditions with wide viewing angle

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and shall be readable from a distance of not less than 16 feet. The console shall display at least six digits per axis simultaneously. The display console shall also provide keypad entry, memory, absolute/incremental counting, data preset, power failure indicator, inch/metric conversion, and absolute zero recall. The electronic display console shall be installed to be readily accessible and easily read from the operator's normal work station. The console shall be mounted to withstand shock and vibration generated by the machine. Each connecting cable shall be armored and shall be securely fastened to the structure of the machine. Location of the installed cables shall prevent any interference with normal machine operation. Electronic display consoles shall be easily and quickly interchangeable between like machines procured under the same contract.

3.2.2 Reclaimed materials. The grinding machine may contain reclaimed materials to the maximum extent possible provided such materials will not jeopardize the intended use, performance, or design life of the machine. Reclaimed materials shall have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used, rebuilt or remanufactured products are allowed under this specification.

3.2.3 Energy efficiency. The grinding machine and its components that directly consume energy in normal operation shall be designed and constructed for energy efficiency as governed by the developments available within the industry.

3.2.4 Controls. All mechanical, hydraulic, and pneumatic operating controls shall be located convenient to the operator's work station(s).

3.2.5 Safety and health requirements. All parts, components, mechanisms, and assemblies furnished on the machine, whether or not specifically required herein, shall comply with all of the requirements of OSHA 29 CFR 1910 and ANSI B11.9. If a conflict arises between 29 CFR 1910 and ANSI B11.9, 29 CFR 1910 shall apply. Additional safety and health requirements shall be as specified (see 6.2) . Covers, guards, or other safety devices normally furnished as standard on the manufacturer's commercial grinding machine supplied to the commercial market shall be provided for the point of operation and all other parts of the machine that present safety hazards. If point-of-operation guarding is influenced by workpiece size and configuration, the procuring activity shall provide part design data to the supplier of the machine (see 6.2). The supplier of the machine shall provide the special guarding which shall meet OSHA 29 CFR 1910.

3.2.6 Mercury restrictions. The grinding machine shall not contain mercury or mercury compounds nor be exposed to free mercury during manufacture.

3.2.7 Asbestos restrictions. Asbestos and materials containing asbestos shall not be used on or in the machine.

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3.2.8 Environmental protection. During operation, service, transportation, or storage of the machine, it shall not emit materials hazardous to the ecological system as prohibited by Federal, state, or local statutes in effect at the point of installation.

3.2.9 Lubrication. An automatic lubricating system shall be provided for all parts and mechanisms requiring a continuous flow of lubricating oil. Means for determining oil level in each reservoir shall be by an oil level sight gauge, a low oil level warning light, or a similar device. Mechanisms requiring periodic manual lubrication shall be readily accessible for servicing. Each recirculating system shall include a cleanable or replaceable filter.

3.2.10 Interchangeability. All parts subject to replacement shall be manufactured to definite dimensions and tolerances. Replacement parts except for hand-fitted parts such as ways shall be interchangeable without requiring modification. Tooling and accessories normally supplied as standard items shall also be considered replacement parts. Replacement parts shall be domestically available.

3.3 Construction. The grinding machine shall be constructed of parts which are new, without defects, and free of repairs. The structure shall withstand all forces encountered during operation of the machine to its maximum rating and capacity without permanent distortion or failure.

3.3.1 Castings and forgings. All castings and forgings shall be free of defects, scale, and mismatching, No processes such as welding, peening, plugging, or filling with solder or paste shall be used for reclaiming any defective part. Such processes may be used only for enhancing surface finish and appearance.

3.3.2 Fastening device. All fasteners shall be installed to prevent change of tightness. Fastening devices subject to removal or adjustment shall not be permanently installed.

3.3.3 Surfaces. All surfaces shall be clean and free of harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. Except as otherwise specified herein, the condition and finish of all surfaces shall be in accordance with the manufacturer's commercial practice.

3.3.4 Welding, brazing, or soldering. Welding, brazing, or soldering shall be employed only where specified in the original design. None of these operations shall be -employed as a repair measure for any defective part.

3.3.5 Painting. Unless otherwise specified (see 6.2), the grinding machine shall be painted in accordance with the manufacturer's commercial practice and color.

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3.3.6 Threads. All threaded parts used on the machine and its related attachments and accessories shall conform to FED-STD-H28 and the applicable "Detailed Standard" section referenced therein.

3.3*7 Drive train gears. All gears of spindle and axis drive trains shall conform to or exceed all provisions of AGMA 390.03a, AGMA 2000.A88, and AGMA 2001.B88 for the inch-pound system of measurement or ANSI/ISO 54 for the metric system of measurement. The gears shall be manufactured of a suitable ferrous material. The gears shall be of proper width and size to transmit full-rated torque and horsepower throughout the speed ranges without failure for the expected service life of the machine. Gears in the drive train shall be hardened and ground steel. Working surface hardness shall be not less than Rockwell C-48.

3.3.8 Electromagnetic interference control. When specified (see 6.2), equipment furnished under this specification shall comply with MIL-STD-461. The 'equipment and subsystems class and the emission 'and susceptibility requirements shall be as specified.

3.4 Components. The grinding machine shall consist of not less than a bed, a traversing table, a swivel table, a grinding wheel head assembly, a head-stock, a footstock, drive mechanisms, a coolant system, a hydraulic system, and an electrical system.

3.4.1 Base. The base shall be constructed to minimize distortion, vibration,and deflection. The base shall support or house all the major components and their related sub-assemblies listed in 3.4. The hydraulic and coolant reservoirs may be integral with the base or separate external units. Access door(s) or removable plate(s) shall be provided for servicing components housed within the base. Guideways shall be provided to accommodate a reciprocating worktable and to provide for cross-axis movement of the grinding wheel assembly. Worktable guideways shall be of such length that the center of gravity of the headstock and footstock is over the bed at all times during normal grinding operations. Style A machines shall have wheelhead guideways which are at a right angle to the table longitudinal guideways. Style B machines shall have wheelhead guideways which are at either 30 degrees, as specified (see 6.2), to the table longitudinal guideways. Guideways shall be provided with protective covers. Leveling screws and floor mounts shall be provided around the periphery of the base.

3.4.2 Headstock. The headstock shall be longitudinally adjustable along the swivel table. Means for positive clamping of the headstock to the table shall be provided. Unless otherwise specified (see 6.2), the headstock shall be constructed for dead center grinding. The headstock shall contain an internal taper conforming to ANSI B5.10 designed to receive a work center. Headstock spindle taper size and type shall be as specified (see 6.2). A drive plate complete with drive pin, which revolves around the work center shall be provided. When a live spindle headstock arrangement is required (see 6.2), the spindle shall be constructed to receive a driving plate and a work center at one end and a driving mechanism at the other. The headstock spindle drive assembly shall be constructed to minimize the generation and transmis-

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sion of vibration. The spindle nose shall conform to ANSI B5.9, spindle nose taper to ANSI B5.10, and unless otherwise specified (see 6.2), taper number and type shall be the manufacturer's standard. A driving plate secured to the spindle shall be furnished. The headstock spindle shall be supported by bearings to minimize axial and radial play. Means for lubricating the spindle shall be provided unless sealed-for-life type bearings are provided. The driving plate bearing shall be protected to prevent the entrance of grit, water or other foreign material to the bearing surface.

3.4.3 Footstock. The footstock shall be longitudinally adjustable along the swivel table. "Means for clamping after adjustment shall be provided. The footstock spindle shall be constructed to retract and extend in a spindle mount. A handwheel, lever, or other means shall be provided for adjusting spindle movement. All moving parts shall be shielded against abrasive grit and coolant. The footstock spindle shall have an internal taper identical in number and type with that of the headstock spindle. The spring tension of the footstock shall be adjustable to ensure the proper center pressure for the weight and size of the workpiece.

3.4.4 Table. The table shall support a swivel table with its components and be arranged to traverse longitudinally on guideways provided by the bed. Unless otherwise specified (see 6.2), power traverse shall be provided with means for manual control. Table traverse shall be fully adjustable with means to prevent damage due to table overtravel. The table reversing mechanism shall operate within the accuracies specified in table I. The table shall have an adjustable dwell control for interrupting table movement at table reversals. Table dwell shall be infinitely adjustable at each end of the stroke.

3.4.5 Swivel table. The swivel table shall be mounted on the traversing table. The swivel table shall be capable of swiveling in the horizontal plane for grinding included angles and tapers. A scale to depict taper, graduated inches per foot and degrees of angle, shall be provided. "Means shall be provided for clamping the swivel table at the desired position. When T-slots are provided for longitudinal adjustment of the headstock and footstock, they shall conform to ANSI B5.1.

3.4.6 Grinding wheelhead assembly. The grinding wheelhead assembly shall include a wheel head housing, spindle, and spindle driving mechanism. The assembly shall operate at a speed necessary to provide a grinding wheel speed within the range of 5000 to 6500 surface feet per minute. The grinding wheel spindle shall be supported by bearings to minimize axial and radial play and shall be dynamically balanced. The spindle shall be constructed to accept a driving mechanism at one end and a grinding wheel at the other end. The spindle shall be mounted in precision anti-friction bearings with continuous lubrication or permanently lubricated sealed for life. The assembly shall be mounted on the guideways provided by the bed and shall traverse crossaxially to the table 90 degrees for style A machines, and either 30 or 45 degrees as specified under 3.4.1 for style B machines. The assembly shall have both automatic and manual movement for infeed of the grinding wheel.

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3.4.6.1 Automatic feed. Each machine shall be designed with automatic infeed of the abrasive wheel for traverse grinding operations. Selectors shall be provided for controlling infeed increments at desired reversal(s) and for limiting the total infeed for the overall grinding cycle. Unless otherwise specified (see 6.2), the wheel feed mechanism shall be actuated upon each or alternate reversals of the table's longitudinal movement, at the choice of the operator.

3.4.6.2 Manual infeed. A handwheel or other manually operated device shall be furnished for feeding the grinding wheel into the work. The hand feed shall have a coarse and fine feed with a graduated indicator/dial.

3.4.7 Automatic cycle. Unless otherwise specified (see 6.2), the grinding machine shall have an automatic cycle for plunge and traverse grinding. After set up for automatic cycle, the operator shall only be required to load the workpiece, start the cycle by a single control and unload the workpiece. The automatic cycle shall include, but not be limited to, the following functions:

3.4.7.1 Plunge grinding.

a. Wheel advances at fast travel, headstock rotation and coolant flow begin.

b. Feed reduces as wheel nears work.

c. Coarse feed engages (rate shall be adjustable or pre-programmed).

d. Fine feed engages as work nears finished size (rate shall be adjustable or pre-programmed).

e. Sparkout: Pre-set positive stop for cross feed starts the automatic spark timing mechanism.

f. Wheel slide retracts rapidly, feed mechanism is reset to original position, headstock rotation stops, coolant flow stops.

3,4.7.2 Traverse winding.

a. Grinding wheel is manually positioned.

b. Power table traverse is actuated, headstock rotation and coolant flow begins.

c. Pre-selected feed engages, wheel feeds at each or alternate table reversals as selected.

d. Sparkout: Pre-set positive stop for cross feed starts the automatic spark timing mechanism.

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e. Power table traverse stops, headstock rotation stops, coolant flow stops, wheel slide retracts, feed mechanism is reset to original position.

3.4.8 Drive system. Unless otherwise specified (see 6.2), the drive system for both the headstock and the grinding wheelhead assembly shall be powered by an electric motor with belt and pulley arrangement. Means shall be provided for adjusting belt tension unless belt tension is maintained automatically.

3.4.9 Diamond wheel dress Each grinding machine shall be equipped with a diamond wheel dresser arranged to dress the entire periphery of the grinding wheel . The dresser shall be complete with a commercial type diamond not less than one (1) carat in total weight.

3.4.10 Dust extractor. When specified (see 6.2), a dust extractor and all necessary connections shall be provided to be used during dry grinding. The dust extractor shall be connected to the machine's electrical system with start/stop pushbuttons conveniently located on the control panel.

3.4.11 Coolant system. The coolant system shall include a sump or reservoir, a power driven pump, and all necessary piping. The sump or reservoir shall have sufficient capacity to permit full flow of coolant. The system shall have means for draining and cleaning and shall include a baffle or a strainer which is easily removable for cleaning. Means shall be provided to permit the operator to direct and control the amount of coolant to the wheel over the entire work area.

3.4.12 Hydraulic system. The hydraulic system shall conform to or exceed the requirements of JIC H-1. The hydraulic system shall be complete, including all pumps, valves, piping, cylinders, and pressure controls. Overpressure protection shall be provided in the high pressure line to prevent damage to components. A filter system shall be provided to ensure delivery of clean fluid. The hydraulic reservoir shall have means for determining fluid level, a provision for draining, and a cleanout access plate if the reservoir is not removable.

3.4.13 Electrical systerm. The electrical system shall be wired to conform to NFPA 79. Each machine shall be wired to draw all of its electrical power from a single circuit. Unless otherwise specified (see 6.2), the machine shall be designed for operation on 3-phase, 60-hertz (Hz), dual 230/460-voltage. Unless otherwise specified (see 6.2), the machine shall be initially wired for operation on 460 volts.

3.4.13.1 Motors. Motors shall meet the requirements of 3.4.13. Motors shall be rated for continuous duty and shall have ball or roller bearings of the sealed and permanently lubricated type. Unless otherwise specified (see 6.2), alternating current (AC) motors shall be designed to operate on 60-Hz. The motor enclosure shall meet the requirements for a dripproof enclosure. All motors shall conform to the requirements of NEMA MG-1.

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3.4.13.2 Worklight. Unless otherwise specified (see 6.2), a worklight shall be provided. The light shall have a protective shield, an on-off switch, and shall be adjustable to reflect maximum light on the work area. If a built-in lighting system is provided, the adjustable light described above shall not be required.

3.4.13.3 Hour meter. When specified (see 6.2), the grinding machine shall be equipped with an hour meter. The hour meter shall be installed to display accumulated operating time of the main drive motor. The meter shall be of the nonresetting type and shall have a range of 0 to 99,999 hours in increments of not greater than 1 hour. The meter shall be sealed to prevent the entrance of dust and moisture and shall be mounted to withstand shock and vibration generated by the machine.

3.5 Size and capacity. Unless otherwise specified (see 6.2), the size and capacity of the grinding machine shall meet the requirements of table I for the machine size specified. Machine size and capacity shall be not less than the stated requirements of the size ordered. When a range is given, the required performance shall be from the stated minimum or less to the stated maximum or greater. Unless otherwise specified (see 6.2), machine physical size shall be not greater than the requirements of the next larger size shown in table 1. When the largest size machine is ordered, the size of the machine offered shall not exceed the stated physical size requirements by more than 10 percent.

3.6 Alignment tolerances. Each grinding machine shall meet the alignment tolerances specified in table II.

3.7 Performance. The machine shall be capable of performing the tests in 4.8.4,

3.7.1 Additional Performance requirements. Performance features required in addition to, or in lieu of, the requirements herein shall be as specified and fully described (see 6.2).

3.8 Standard equipment: In addition to equipment normally provided with the manufacturer's commercial machine, standard equipment shall include a faceplate and a chuck sized compatible with the machine size required, one each headstock and footstock centers, and a minimum of two grinding wheels, one suitable for grinding mild steel and one suitable for grinding hardened tool steel.

3.9 Optional equipment. Optional equipment shall be furnished as specified and fully described (see 6.2). The optional equipment provided shall be functional-without requiring modification of the equipment or machine.

3.10 Repair parts. Required repair parts shall be furnished as specified (see 6.2).

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3.11 Marking on instruments, control panels, charts, and plates. Al 1 words on instruments, control panels, charts, and plates shall be in the English language. Characters shall be permanently marked in boldface on a contrasting background. All plates shall be corrosion resistant. ,

3.11.1 Lubrication plate. When specified (see 6.2) a lubrication plate shall be attached to each machine. Unless otherwise specified (see 6.2), the information provided on the plate shall be as listed below.

Points of lubricant application
Type of lubricant
Servicing interval
Military or Federal specification for each lubricant (if applicable)

3.11.2 Nameplate. A nameplate shall be securely attached to each machine. Unless otherwise specified (see 6.2), the nameplate shall contain the information listed below.

Nomenclature
Manufacturer's name
Manufacturer's model designation
Manufacturer's serial number
Power input (volts, total amps, phase, frequency)
Amp rating at largest motor
Short circuit/over-current protection rating
Contract number or order number
National Stock Number or Plant Equipment Code
Date of manufacture

3.11.3 Hydraulic fluid requirements plate. When specified (see 6.2), a hydraulic fluid requirements plate shall be attached to the machine at the hydraulic reservoir fill point. Unless otherwise specified (see 6.2), the plate shall contain the following information.

Type of fluid (SAE or military or Federal specification number)
Type of filter

3.12 Workmanship. of the grinding machine and its accessories shall meet all requirements specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. , Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

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4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submissions of known defective material, either indicated or actual, nor does it commit the government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

c. Acceptance test (see 4.6).

4.3 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be performed in accordance with the test conditions specified in 4.8.1.

4.4 First article inspection. When a first article inspection is required, it shall be applied to the first article submitted in accordance with 3.1. Unless otherwise specified (see 6.2), first article inspection shall consist of the examination in 4.7 and all tests in 4.8. The machine shall pass the first article examination and all tests to be accepted.

4.5 Quality conformance inspection. Quality conformance inspection shall be applied to each machine prior to being offered for acceptance under the contract. Unless otherwise specified (see 6.2), quality conformance inspection shall consist of the examination in 4.7, the tests in 4.8, and the inspection in 4.9. The machine shall pass the examination, all tests, and the inspection to be accepted.

4.6 Acceptance test. Acceptance test shall be performed on the machine to ensure conformance with this specfication. Unless otherwise specified (see 6.2), the acceptance test shall be performed only after the machine is installed at its final location. The acceptance test shall consist of the examination in 4.7 and all tests in 4,8. The machine shall pass the examination and all tests to be accepted.

4.7 Examination. Each machine shall be examined to determine compliance with all requirements of this specification.

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4.8 Tests.

4.8.1 Test conditions. All tests shall be performed in an indoor facility with ambient conditions if 41 degrees to 104 degrees Fahrenheit and 20 percent to 95 percent relative humidity.

4.8.2 Operational test. The machine shall be operated at no load for not less than 2 hours. Proper operation of all controls, motors, adjusting mechanisms, and accessories shall be verified during the trial period.

4.8.3 Alignment tolerance test. Each machine shall be tested to determine compliance with the alignment tolerance requirements of 3.6. The alignment tolerance test shall be performed prior to and subsequent to the performance test given in 4.8.4. Any machine adjustments required to comply with 3.6 during or after the performance test shall be cause for rejection.

4.8.4 Performance tests. Each style A machine shall grind a 12 inch long cylindrical workpiece round within 0.0001 inch of a desired diameter, with not more than 0.0001 inch taper per foot, and a surface roughness not more than 5 micro-inches. Each style B machine shall plunge grind a similar workpiece with a radial shoulder on one end to the above tolerances. In addition, the axial runout of the ground shoulder shall not exceed 0.0001 inch TIR. The workpiece diameter, material, and methods used to obtain the above designated requirements shall be at the contractor's option. All performance tests shall be completed without making any adjustments to or performing any maintenance on the machine.

4.8.4.1 Additional performance tests. Performance tests required in addition to or in lieu of the requirements given in 4.8.4 shall be as specified and fully described (see 6.2).

4.8.5 Electromagnetic interference control tests. Equipment requiring electromagnetic interference control shall be tested for compliance with 3.3.8 using the procedures- given in MIL-STD-462.

4.9 Packaging inspection. Packaging of each item shall be inspected to determine compliance with the requirements of section 5.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-18058.

6. NOTES

6.1 Intended use. The grinding machines covered by this specification are intended for use in shops where precision grinding to uniform tolerances for size, roundness, and finish on a wide variety of materials is required.

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6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification,

b. Type, class, and size required (see 1.2).

c. Issue of DODISS to be cited in the solicitation and, if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).

d. First article, if required (see 3.1).

e. System of units for measuring and indicating devices (see 3.2.1).

f. Additional safety and health requirements,if required (see 3.2.5).

g. Specify configuration of additional point-of-operation guarding, if required (see 3.2.5).

h. Painting, if different (see 3.3.5).

i. If electromagnetic interference control is required, specify the equipment and subsystem class and the emission and susceptibility required (see 3.3.8).

j. For style B machines, specify whether 30 or 45 degree guideways (see 3.4.1).

k. Headstock center, indifferent (see 3.4.2).

l. Specify headstock spinalle taper size and type required (see 3.4.2).

m. Specify if live spindle headstock arrangement is required (see 3.4.2).

n. Specify taper number and type, if different (see 3.4.2).

o. Power traverse, if not required (see 3.4.4).

p. Wheel feed actuation, if different (see 3.4.6.1).

q. Automatic grinding cycle, if different (see 3.4.7).

r. Drive system, indifferent (see 3.4.8).

s. Dust extractor, if required (see 3.4.10).

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t. Electrical system, if different (see 3.4.13).

u. Initial voltage, if different (see 3.4.13).

v* Motors, if different (see 3.4.13.1).

w. Worklight, if not required (see 3.4.13.2).

x. Hour meter, if required (see 3.4.13.3).

y" Size and capacity, indifferent (see 3.5).

z. Additional performance features, if required (see 3.7.1).

aa. Specify and fully describe optional equipment, if required (see 3.9).

bb. Specify applicable repair parts, if required (see 3.10).

cc. Lubrication plate, if required (see 3.11.1).

dd. Information to be provided on lubrication plate, if different (see 3.11.1).

ee. Nameplate, if different (see 3.11.2).

ff. Hydraulic plate, if required (see 3.11.3).

gg. Information to be provided on hydraulic plate, if different (see 3.11.3).

hh. Inspection conditions, if different (see 4.3).

ii. First article inspection, if different (see 4.4).

jj. Quality conformance inspection, if different (see 4.5).

kk. Acceptance test, if different (see 4.6).

11. Additional performance test, if required (see 4.8.4.1).

6.3 consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DIDs) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DIDs are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DOD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

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Reference Paragraph	DID Number	DID Title	Suggested Tailoring
.	.	.	.
.	.	.	.

The above DIDs were those cleared as of the date of this specification. The current issue of DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DIDs are cited on the DD Form 1423.

6.3.1 Technical data. It should be specified in the remarks section of the DD Form 1423 that all technical data ordered shall be written in the English language.

6.4 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a first article sample, a first production item, or a standard production item from the contractor's current inventory, and the number of items to be tested as specified in 4.4. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.5 Definitions.

6.5.1 Inch-pound units. Inch-pound units are a system of measures based on the yard and pound commonly used in the United States of America and defined by the National Bureau of Standards. Inch-pound units having the same names in other countries may differ in magnitude.

6.5.2 Metric units. Metric units are a system of basic measures defined by the International System of Units based on "Le Systeme International D'Unites (SI)," of the International Bureau of Weights and Measures. These units are described in ASTM E 380 and IEEE 268 (MIL-STD-961).

6.6 Measurement system. In this specification, all measurements, dimensions, sizes, and capacities are given in inch-pound units. These measurements may be converted to metric units through the use of the conversion factors and methods specified in FED-STD-376.

6.7 Changes from previous issues. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

6.8 Safety and health requirements. OSHA 29 CFR 1910 limits only the total hazard level (noise, radiation, electromagnetic emissions, noxious vapors, air

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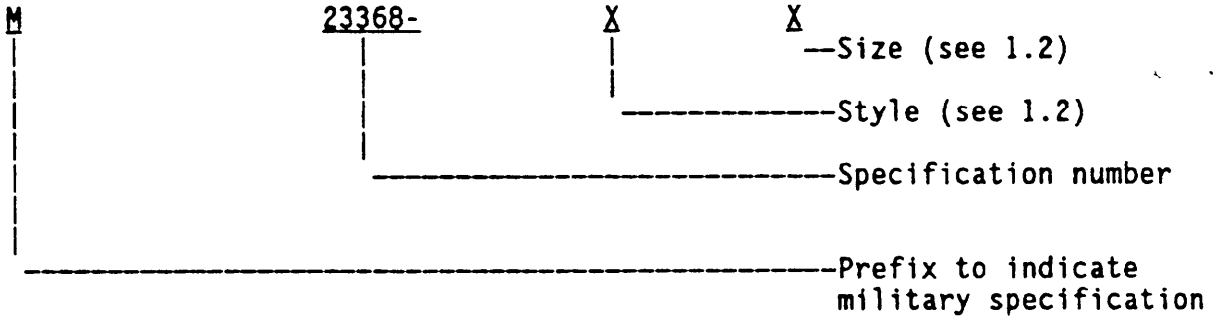
contaminants, and heat) of the environment in which a machine will operate. It does not limit the hazard level of individual machines in an operating environment. The procuring activity should analyze the existing hazard level in the proposed operating environment and specify additional requirements necessary to integrate this new machine into its future environment.

6.9 Training. Training required by the procuring activity should be provided as specified in the contract.

6.10 Warranty. Warranty requirements should be as specified by the procuring activity in the contract.

6.11 Inspection location. The contractor should identify in his response to the solicitation the location where inspection and tests are to be performed.

6.11 Part or Identifying Number (PIN). The PIN to be used for machines acquired to this specification are created as follows:



6.12 Subject term (keyword) listing.

Automatic
Cylindrical
External
Grinding
Machine tool
Plunge grinding
Traverse grinding

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TABLE I - Sizes and Capacities Style A Machines

CHARACTERISTIC / SIZES	412	612	812	1018	1218	1418	1624	1824	2424
Optional sizes - in 6" increments (same as rated swing)	418-430	618-636	818-836	1024-1048	1224-1248	1424-1448	1630-1648	1830-1848	2430-2448
in 12" increments (same as rated swing)				1072-1096	1272-1296	1472-14120	1672-16120	1872-18168	2472-24168
Swing over table, inches	4	6	8	10	12	14	16	18	24
Distance between centers, inches	12	12	12	18	18	18	24	24	24
Optional distances between centers - 6" increments	18-30	18-36	18-36	24-48	24-48	24-48	30-48	30-48	30-48
24" increments	---	---	---	72-96	72-96	72-120	72-120	72-120	72-168
Table travel, inches	15	15	15	21	21	21	27	27	27
Swivel table, amount of swivel angle in degrees									
Toward wheelhead	11	12	12	12	12	12	12	12	--
Away from wheelhead	2	3	3	3	3	5	5	5	--
Swivel table, graduated taper per foot, inches									
Toward wheelhead	4-3/4	5	5	5	5	5	5	5	3-3/8
Away from wheelhead	1	1	1	1	1	2	2	2	1-1/4
*Grinding wheelhead:									
Grinding wheel (outside diameter by face width by hole diameter, inches)	16x1x8	20x2x12	24x2x12	24x2x12	24x2x12	24x2x12	30x3x12	30x3x13	30x3x12
Spindle speed, RPM	1550	1240	1030	1030	1030	830	830	830	830
*Headstock:									
Speeds, number	4	4	4	4	4	infinite	infinite	infinite	infinite
Speed range, RPM	175-800	97-395	97 - 395	66 - 268	66 to 268	63 to 212	40 to 144	20 to 72	11 to 44
*Table traverse:									
Grinding rates, number	<---------- infinite ---------->								
Minimum movement of table necessary for automatic reversal, inches, not more than	1/16	3/32	3/32	3/32	3/32	3/32	1/2	1/2	1/2
Range of grinding rates, inches per minute	3 to 280	3 to 240	3 to 240	3 to 240	3 to 240	3 to 220	3 to 120	3 to 120	3 to 120
Accuracy of table at all traverse rates, inches, not more than	.001	.004	.004	.004	.004	.004	.005	.005	.005
Table dwell adjustment, seconds	0 to 5	0 to 5	0 to 5	0 to 5	0 o 5	0 to 5	0 to 15	0 to 15	0 to 15
Hand traverse/rate per revolution of handwheel, inches									
Slow, not more than	.010	.200	.200	.200	.200	.200	.200	.200	.200
Rapid, not less than	1	15/16	15/16	15/16	15/16	15/16	1	1	1
*Infeed adjustments:									
Wheelhead movement per revolution of infeed handwheel, inches, not more than	.050	.050	.050	.050	.050	.050	.050	.050	.050
Infeed increment, diameter reduction, inches, not more than	.00003	.00003	.00003	.00003	.00003	.00003	.00003	.00003	.00003
Automatic infeed at table reversal, diameter reduction, range, inches	<---------- .0002 - .005 ---------->								
Total adjustment of wheelhead by hand	5-1/2	7-1/2	9-1/2	11-1/2	13-1/2	15-1/2	17-1/2	19-1/2	25-1/2
*Motors, horsepower:									
Wheelhead	3	7-1/2	7-1/2	7-1/2	7-1/2	15	20	20	20
Headstock	1/3	1/2	1/2	3/4	3/4	1	1-1/2	1-1/2	3
Hydraulic pump	3/4	1	1	1	1	1-1/2	1-1/2	1-1/2	2
coolant pump	1/4	1/4	1/4	1/4	1/4	1/4	1/4	1/4	1/3

* Unless otherwise specified (see 3.5)

NOTE: 1 - Machine sizes and capacities shall not be less than the stated requirements of the size specified in 6.2.1, nor shall they be greater than the next larger size shown in table 1 above. Provided the largest size and capacity is specified, it may not exceed the stated requirements by more than 10 percent. When a range is shown, the required performance is from the minimum or less to the maximum or greater.

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TABLE I - Sizes and Capacities Style B Machines

CHARACTERISTIC	REQUIREMENT								
SIZES	412	612	812	1018	1218	1418	1624	1824	2424
Optional sizes - in 6" increments (same as rated swing)	418-430	618-636	818-836	1024-1048	1224-1248	1424-1448	1630-1648	1830-1848	2430-2448
in 12" increments (same as rated swing)				1072-1096	1272-1296	1472-14120	1672-16120	1872-18168	2472-24168
Swing over table, inches	4	6	8	10	12	14	16	18	24
Distance between centers, inches	12	12	12	18	18	18	24	24	24
Optional distances between centers - 6" increments	18-30	18-36	18-36	24-48	24-48	24-48	30-48	30-48	30-48
24" increments				72-96	72-96	72-120	72-120	72-120	72-168
Table travel, inches	15	15	15	21	21	21	27	27	27
Swivel table, amount of swivel angle in degrees									
Toward wheelhead	411	12	12	12	12	12	12	12	--
Away from wheelhead	2	3	3	3	3	5	5	5	--
Swivel table, graduated taper per foot, inches									
Toward wheelhead	4	5	5	5	5	5	5	5	3-3/8
Away from wheelhead	1	1	1	1	1	2	2	2	1-1/4
*Grinding wheelhead:									
Grinding wheel (outside diameter by face width by hole diameter, inches)	20x2x12	24x2x12	24x2x12	24x2x12	24x2x12	30x3x12	30x3x12	30x3x13	30x3x12
*Spindle speed, RPM	1550	1240	1030	1030	1030	830	830	830	830
*Headstock:									
Speeds, number	4	4	4	4	4	infinite	infinite	infinite	infinite
Speed range, RPM	175 - 800	97 - 395	97 - 395	66 to 268	66 to 268	63 to 212	40 to 144	20 to 72	11 to 44
*Table traverse:									
Grinding rates, number	<----------infinite---------->								
Minimum movement of table necessary for automatic reversal, inches, not more than	1/16	3/32	3/32	3/32	3/32	3/32	1/2	1/2	1/2
Range of grinding rates, inches per minute	3 to 280	3 to 240	3 to 240	3 to 240	3 to 240	3 to 220	3 to 120	3 to 120	3 to 120
Accuracy of table at all traverse rates, inches, not more than	.001	.004	.004	.004	.004	.004	.005	.005	.005
Table dwell adjustment, seconds	0 to 5	0 to 5	0 to 5	0 to 5	0 to 5	0 to 5	0 to 15	0 to 15	0 to 15
Hand traverse/rate per revolution of handwheel, inches									
Slow, not more than	.010	.200	.200	.200	.200	.200	.200	.200	.200
Rapid, not less than	1	15/16	15/16	15/16	15/16	15/16	1	1	1
*Infeed adjustments:									
Wheelhead movement per revolution of infeed handwheel, inches, not more than	.050	.050	.050	.050	.050	.050	.050	.050	.050
Infeed increment, diameter reduction, inches, not more than	.00003	.00003	.00003	.00003	.00003	.00003	.00003	.00003	.00003
Automatic infeed at table reversal, diameter reduction, range, inches	<----------.0002 - .005---------->								
Total adjustment of wheelhead by hand, inches	6	8	10	12	14	16	18	20	25-1/2
*Motors, horsepower:									
Wheelhead	3	7-1/2	7-1/2	7-1/2	15	15	20	20	20
Headstock	1/3	1/2	1/2	3/4	3/4	1	1-1/2	1-1/2	3
Hydraulic pump	3/4	1	1	1	1	1-1/2	1-1/2	1-1/2	2
coolant pump	1/4	1/4	1/4	1/4	1/4	1/4	1/4	1/4	1/3

* Unless otherwise specified (see 3.5)

NOTE: 1 - Machine sizes and capacities shall not be less than the stated requirements of the size specified in 6.2.1, nor shall they be greater than the next larger size shown in table 1 above. Provided the largest size and capacity is specified, it may not exceed the stated requirements by more than 10 percent. When a range is shown, the required performance is from the minimum or less to the maximum or greater.

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TABLE II. Alignment Tolerances

Tests to be applied	Maximum Allowable deviation
Bed and table:	
Bed straight in longitudinal direction	0.00025" per ft.
Bed flat or level in transverse direction, no twist permitted	0.00025" per ft.
Guideways of headstock and footstock parallel with table movement	0.00012" per ft.
Headstock and spindle:	
Axis of headstock parallel with table movement in vertical plane (rising toward free end the mandrel)	0 to 0.0005" per ft.
Axis of headstock parallel with table movement in horizontal plane (free end of mandrel inclined toward grinding wheel)	0 to 0.0005" per ft.
*Center point for true running	0.0002" TIR
*Taper of work spindle runs true:	
(1) Nearest spindle nose	0.0002" TIR
(2) At a distance of 12 inches	0.0004" TIR
Footstock:	
Sleeve parallel with table movement in vertical plane (front end rising only)	0 to 0.001" per ft.
Sleeve parallel with table movement in horizontal plane (free end of mandrel inclined toward the grinding wheel)	0 to 0.001" per ft.
Taper in sleeve parallel with table movement in vertical plane (free end of mandrel rising)	0 to 0.0005" per ft.
Axis of centers (mandrel between centers) parallel with table movement in vertical plane (rising toward footstock end)	0 to 0.0005" per 12 in.
Grinding wheel spindle:	
Taper of spindle running true	0.0005" TIR
Spindle for axial slip measured at two points, displaced by 180 degrees	0.001"average TIR
Axis of wheel spindle parallel with table movement in vertical plane (rising toward free end of mandrel)	0 to 0.0012" per ft.
Axis of wheel spindle parallel with table movement in horizontal plane (free end of mandrel inclined toward table)	0 to 0.0012" per ft.
Axis of wheel spindle and headstock spindle at same height with respect to swivel plate	0.004"
Rise and fall of wheel spindle in its infeed movement	0.002" in total length of infeed motion
Infeed motion of wheelslide square with bedways (upper table set for cylindrical grinding by means of a dial gage) (style A)	0.001" on total length of infeed motion
Quick approach to the work, repeats accurately to grinding position (6 repetitions), fine infeed	0.0001"
Working accuracy of machine:	
Machine grinds round up to 3" diameter	0.0001"
From 3" to 8" diameter	0.0002"
Over 8" diameter	0.0004"
Machine grinds cylindrically without the use of steady rests (convex only):	
Shafts, up to 1-1/2" dia. by 10" length	0.0002"
Shafts, up to 2" dia. by 20" length	0.0003"
Shafts, up to 3" dia. by 40" length	0.0006"

*Applicable to live spindle headstock only.

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Army - AL
Navy - SH
Air Force - 99

Review activities:
DLA - GS
Air Force - 84

User activities:
Navy - AS, OS

Preparing activity:
DLA - IP

Project (3415-0172)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-G-23368D	Grinding Machines, Cylindrical, External, Plain, Power.

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FOR ALONG THIS LINE.)

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